

BRAD BURCH

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SUMMARY

Full-stack software engineer with 7 years shipping production systems. Most recently built and run a booking, payments, and Claude agent platform (MCP, OAuth 2.1 + PKCE, Stripe Connect) on Cloudflare Workers and D1 with TypeScript and Hono, with 4+ outside pet-sitting businesses now on the platform.

SKILLS

Languages & Frameworks: TypeScript, JavaScript, Python, Java, Scala, SQL, React 19, Node.js, Hono, Spring, Play

Infrastructure: AWS (DynamoDB, SQS), serverless (Cloudflare Workers, Durable Objects, Queues, Workflows), PostgreSQL, SQLite (D1), S3-compatible object storage (R2), KV, MSSQL, Docker, Kubernetes, REST APIs, webhooks, distributed systems

AI / LLM: Claude (Haiku, Sonnet), Vercel AI SDK, Model Context Protocol (MCP), tool calling, LLM evals, prompt caching, prompt-injection defense, AI cost metering, Claude Code, Cursor

Payments & Security: Stripe (Connect, Billing, Checkout), OAuth 2.1 + PKCE, JWT, SAML 2.0, HMAC webhook verification, AES-GCM, SSRF prevention, multi-tenant isolation

Testing & Delivery: Vitest (Node and Workers runtime), Playwright, contract tests, GitHub Actions, Jenkins, CI/CD, alerting

EXPERIENCE

Software Engineer

2022 - Present

Pawsevation, San Francisco, CA (pawsevation.com) | booking, payments, and AI platform for pet-care businesses

- Built a production TypeScript booking and payments platform (React, Hono, Cloudflare Workers, D1, KV, Workflows) with no origin server, serving 30+ active clients, 60+ bookings/month, and 30+ invoices/month.
- Shipped a Claude Haiku 4.5 booking agent (Vercel AI SDK, 15 tools) on web chat and WhatsApp plus a remote MCP server, which with the client portal moved more than half of bookings to self-serve.
- Cut agent cost 4.9x per eval run by restructuring the system prompt into a byte-stable cached prefix, serving about 95% of eval-suite input tokens from cache.
- Guaranteed each stay is auto-charged at most once by claiming a D1 uniqueness constraint before calling Stripe, verified with 20 concurrent writers on the real D1 runtime.
- Replaced date-ordered payment matching with closest-first allocation, ending a defect class that cleared a \$600 stay with a \$40 payment and flagged a \$2,000 prepayment as delinquent.
- Added change detection to a scheduled calendar sync, eliminating the source of 97% of daily database writes (56,424 of 58,121 against a 100,000/day platform cap).
- Cut per-invoice PDF CPU time from 154 ms to 18 ms after Workers CPU limits cut monthly invoice runs short, adding completion markers with catch-up.
- Added a real-Workers-runtime CI suite after tracing a Claude connector outage to a fetch option Node accepts and Workers rejects, so that bug class now fails the build.

Software Engineer

2022

Front, San Francisco, CA (role eliminated in company-wide layoff)

- Shipped a Node.js REST API enabling account-level email rules for multi-domain enterprise customers, previously configurable only per-inbox.
- Shipped a security upgrade to the core JavaScript email library handling 200K+ outbound messages per month.

Software Engineer

2019 - 2021

Total Brain, San Francisco, CA

- Shipped a Scala backend integration for Apple, Google, and Facebook authentication, contributing to a 60% lift in new sign-ups.
- Migrated user data from MSSQL to DynamoDB with a redesigned NoSQL schema, cutting average API response times by 50%; mentored junior engineers on backend patterns.
- Standardized SAML 2.0 and JWT SSO and built HubSpot and Box integrations, cutting enterprise onboarding from 2+ weeks to ~3 days.
- Served as engineering point of contact on enterprise Customer Success and Sales calls, scoping integrations, demoing to technical and business stakeholders, and debugging live.

Software Engineer

2017 - 2018

Salesforce.com, San Francisco, CA

- Built automated testing frameworks on a developer-productivity team, including a persistent test-state repository (Java, PostgreSQL, EclipseLink).
- Built an OAuth2 security filter for microservice API access control with remote identity validation.
- Built a React management UI that internal engineers used to configure, launch, and inspect microservice test runs, cutting internal test setup time by 40%.
- Led delivery of a config builder and validator used by internal teams, scoping milestones and coordinating partner teams.

EDUCATION

DePauw University, Bachelor of Arts in Computer Science

2017